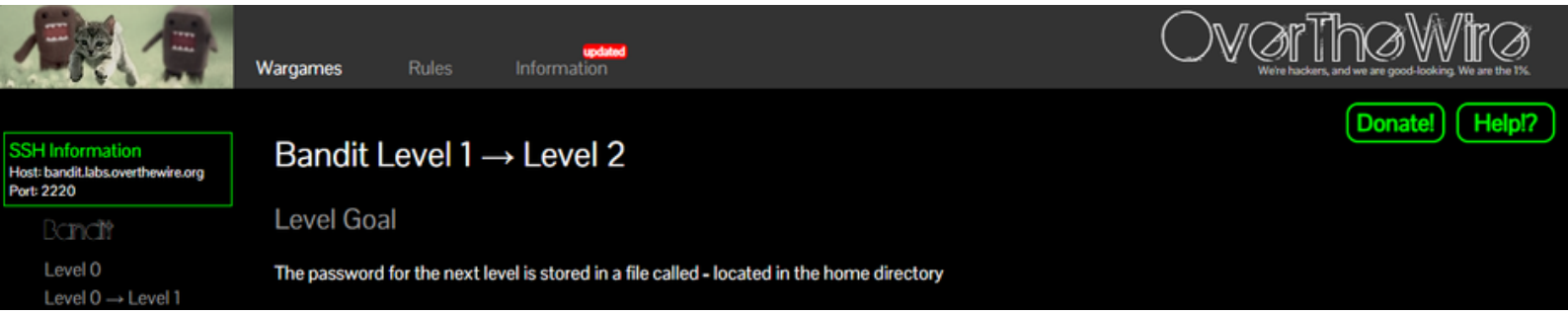

Bandit Level 1 ---> Level 2

1. Connection Details

- **Host:** bandit.labs.overthewire.org
- **Port:** 2220
- **Username:** bandit2
- **Password:** 263JGJPfgU6LtdEvgfWU1XP5yac29mFx (which you got from previous level1)

```
cyres@ubuntu-GUI:~$ ssh bandit2@bandit.labs.overthewire.org -p 2220
                      _                     _ _ _   _
                     | |__   __ _ _ __   __| (_) |_
                     | '_ \ / _` | '_ \ / _` | | __|
                     | |_) | (_| | | | | (_| | | |_
                     |_.__/ \__,_|_| |_|\__,_|_|\__|


                    This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit2@bandit.labs.overthewire.org's password:
```

entered into the **bandit2** level, so lets explore this level

```
  For support, questions or comments, contact us on discord or IRC.

  Enjoy your stay!

bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$
```

Exploration: You used the **ls** (list) command to see what files were in your home directory.

Discovery: You identified a file called “- -spaces in this filename --”

The challenge here is that the filename has **spaces** in it. In Linux, if you just type cat spaces in this filename, the computer thinks you are trying to read 4 different files: one named "spaces", one named "in", and so on.

so lets try to read the file using **cat** command

```
bandit2@bandit:~$ cat --spaces in this filename--
cat: unrecognized option '--spaces'
Try 'cat --help' for more information.
bandit2@bandit:~$
```

When you use **cat --spaces in this filename** in this filename, two separate things go wrong simultaneously in the eyes of the Linux shell:

1. The "Option" Confusion

The shell sees the -- at the start and thinks you are trying to provide a command-line **flag** or **option**. Since cat doesn't have an option named --spaces, it throws the error:

cat: unrecognized option '--spaces'

2. Word Splitting

Because there are spaces and no quotes, the shell "splits" your command into multiple arguments. It thinks you are asking it to do this:

- **Argument 1:** cat (The command)
- **Argument 2:** --spaces (An option/flag)
- **Argument 3:** in (A second file to open)
- **Argument 4:** this (A third file to open)
- **Argument 5:** filename (A fourth file to open)

so lets solve this two problem in one command

To solve both the **"Option Confusion"** (the --) and the **"Word Splitting"** (the spaces) in a single move, you should use the **Path + Quotes** combination.

```
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat --spaces in this filename--
cat: unrecognized option '--spaces'
Try 'cat --help' for more information.
bandit2@bandit:~$ cat "./--spaces in this filename--"
MNk8KNH3Usiio41PRUEoDFPqfxLPlSmx
bandit2@bandit:~$
```

The **./** (Solves the Option Problem): By starting the filename with a path, you force cat to treat the string as a **location** on the disk. Even though the filename starts with dashes, it is no longer the *first* character the command sees, so it doesn't get mistaken for a flag.

The **"** (Solves the Space Problem)**:** Wrapping the whole thing in double quotes forces the shell to see the entire string as **one single filename** instead of five separate words.

```
bandit2@bandit:~$ cat ./--spaces\ in\ this\ filename--
MNk8KNH3Usiio41PRUEoDFPqfxLPlSmx
bandit2@bandit:~$
```

The "Pro" Alternative

If you want to feel like a real power user, you can use **Tab Completion**.

1. Type **cat ./-**
2. Press the **Tab** key.

The terminal will automatically write out the rest, usually looking something like this:

cat ./--spaces\ in\ this\ filename--

(Note: The backslashes \ you see there are another way of telling Linux "the next space is part of the name".)

so now the password for next level is **MNk8KNH3Usiio41PRUEoDFPqfxLPSmx**